

AAKANSHA NARGAWE

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EDUCATION

Indian Institute of Technology (BHU), Varanasi
IDD – Mathematics and Computing

2023 – Present

Jawahar Navodaya Vidyalaya, Barwani
Class XII (CBSE)

2022

TECHNICAL SKILLS

- **Languages:** Python, C, C++, SQL
- **Backend & Web:** FastAPI, React.js, JavaScript, HTML, REST APIs
- **Databases:** PostgreSQL, SQLAlchemy
- **AI/ML:** Gemini API, Prompt Engineering, Document Processing
- **Tools:** Git, VS Code, Node.js, Express
- **Coursework:** Data Structures & Algorithms, Operating Systems, Object-Oriented Programming

PROJECTS

- **LLM-Based Document Intelligence & Knowledge Extraction Pipeline**
 - Developing an **LLM-powered document understanding system** using **Python, FastAPI, PostgreSQL, SQLAlchemy, and Gemini API**.
 - Implemented **context-aware sliding window processing**, semantic segmentation, and hierarchical metadata extraction for long-form documents.
 - Built a scalable backend with **asynchronous REST APIs**, middleware, Pydantic, transactional workflows, and structured logging.
- **Producer-Consumer Synchronization Simulator (C++)**
 - Implemented the classic **Producer-Consumer synchronization problem** using a shared queue-based buffer for inter-process communication.
 - Developed a **menu-driven simulation** handling buffer overflow/underflow and synchronized producer-consumer operations.
- **Novel Activation Function for Deep Learning**
 - Analyzed a **custom activation function** and compared its performance with **ReLU** and **Sigmoid**.
 - Evaluated its impact on **model convergence** and learning behavior across neural network architectures.

ACHIEVEMENTS

- Ranked among the **Top 55 Teams Nationwide** in the **Microsoft AI Unlocked Hackathon 2026: Campus Edition**.
- Shortlisted for the interview round of the **Texas Instruments WiSH Mentorship Programme 2026**.
- Represented **IIT (BHU)** at the **Inter IIT Sports Meet (IIT Bombay, IIT Roorkee)** and secured a **Silver Medal**.
- **Gold Medalist** – Kho Kho, **Spardha'25**, IIT (BHU).
- Secured **1st Position** in the **General Championship**, IIT (BHU).
- Completed the **JavaScript and React.js Bootcamp** conducted by **Google Developer Student Clubs**.

POSITIONS OF RESPONSIBILITY

- Led team coordination and match strategy as **Vice Captain**, securing a **Silver Medal** at **Sangram'25 (IIT Roorkee)**.
- **Coordinated Spardha'25**, managing logistics and operations for **30+ teams** from IITs and colleges.